

ChatGPT Media Coverage Metrics; Initial Examination

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ABSTRACT

This paper presents an overview of coverage of OpenAI's ChatGPT in media outlets from November 2022-March 2023, a comparison to previous media coverage of the chatbot Tay across the same outlets, and a count of ChatGPT media articles pertaining to government legislation and regulation. The New York Times, Wired, Gizmodo, The Globe and Mail, and The Guardian were searched for coverage. Across all five outlets there is an uptick in media coverage surrounding ChatGPT, with total numbers of included articles per month being 0 in November, 39 in December, 68 in January, 104 in February, and 143 in March. Findings exemplify the trend of increased coverage of ChatGPT in media public discourse, which contrasts with previous smaller media coverage of Tay. Examination of headlines and subheadings of included articles reveals minimal coverage (5.7%) dedicated to government legislation of ChatGPT. Future research will evaluate what is being said about ChatGPT within these media outlets.

KEYWORDS

ChatGPT; Media Coverage; Tay; Legislation

INTRODUCTION

Library and Information Science (LIS) should be interested in chatbots, such as ChatGPT, because of the field's long history of analyzing and critiquing information systems, including conversational agents or chatbots (e.g., Rubin et al., 2010). A chatbot is "a computer program that uses AI and natural language processing (NLP) to understand customer questions and automate responses to them, simulating human conversation" (IBM, (n.d)). Shum et al. (2018) note that "perhaps the very first chatbot known publicly" (p. 11) was Eliza, a chatbot created by a MIT researcher in 1966. Over 55 years since the creation of Eliza, on November 30th, 2022, OpenAI released its artificial intelligence (AI) chatbot ChatGPT which quickly gathered media interest. Previous studies have noted that the use of AI in "[...] both the private and public sectors has captured the interest of media and citizens" (Igartua et al., 2022, p. 2), and news coverage "makes trends visible, influences public discourses, contributes to the discursive construction of benefits and risks (Lupton, 2013), and shape technology perceptions (Pentzold et al., 2019)" (Nguyen & Hekman, 2022, p. 1). The media coverage surrounding ChatGPT seems to differ from previously released chatbots. This work seeks to examine coverage of ChatGPT in media discourse, assess if media coverage of ChatGPT differs from past coverage on a previously released chatbot (Tay) and provide a glimpse into the type of topic coverage surrounding ChatGPT. Tay was released on March 23rd, 2016 by Microsoft Corporation and was selected as a comparable chatbot that also created a splash in the media due to its promise in AI conversational technology innovation.

METHODS – DATA COLLECTION

The media outlets of The New York Times, Wired, Gizmodo, and The Guardian were selected for "their international scope and focus on technology [...] and influential voices in global media discourses on technology" (Nguyen & Hekman, 2022, p. 2). This decision parallels Nguyen and Hekman's (2022) study that examined media rhetoric of artificial intelligence across these same four media outlets. The Globe and Mail was also included to represent Canadian media coverage. Across all five media outlets the search string of 'ChatGPT' was used, and each media outlet was searched for articles from the period of November 1st, 2022 to March 31st, 2023. Exclusion criteria were set to omit articles that did not mention ChatGPT within the title, subheading, or first two paragraphs of the text. Additionally, the following retrieved items were excluded across media outlets: quizzes, briefing/summary articles, letters to the editors, applications/tools, quotes of the day, videos, podcasts, advisory weekly letters, investor newsletters, global advisory reports, cartoons, advertisement of products/courses, paid content, call for opinions, and duplicate articles. Apart from including any retrieved hits that mentioned Tay within the title, subheading, or text of the article, the same approach to searching was conducted for collection of media articles published in the year 2016 and pertaining to Tay. We also scanned the headings and subheadings of included articles pertaining to ChatGPT and counted how many of them mentioned government legislation or regulation, omitting mentions of school policies, and concerns about legality of ChatGPT output and its uses (e.g., pertaining to copyright law).

INCREASE IN MEDIA COVERAGE

Figure 1 depicts the number of media articles meeting inclusion criteria per month across all five media outlets (The New York Times, Wired, Gizmodo, The Globe and Mail, and The Guardian).

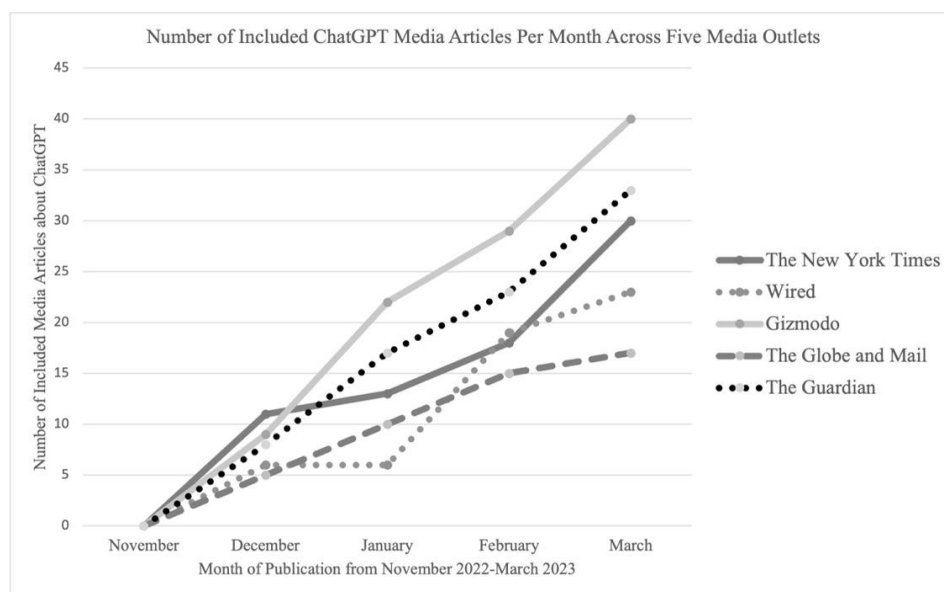


Figure 1. Number of Included ChatGPT-Related Media Articles Per Month Across Five Media Outlets

As clearly visible in the graphic, all five media outlets had a sharp increase in the number of articles pertaining to ChatGPT from November 1st 2022 to March 31st 2023. The average numbers of articles collected from each organization per month were: 0 in November, 7.8 in December, 13.6 in January, 20.8 in February, and 28.6 in March. When combining the total number of included articles across media outlets per month, the total numbers of articles steadily increased from month to month: 0 in November, 39 in December, 68 in January, 104 in February, and 143 in March. In total, 354 articles had substantial coverage of ChatGPT and met our inclusion criteria.

Comparison to Tay Chatbot Media Coverage

A total of 354 included media articles were collected from all five media outlets pertaining to ChatGPT, while Tay's media coverage was about 17 times smaller during the entire year of 2016. Only 21 Tay-related media articles were gathered from the same five outlets. In comparison to Tay, ChatGPT is clearly receiving more media attention akin of a hype. While AI may "capture the interest of media and citizens" (Igartua et al., 2022, p. 2), there appears to be a different level of media coverage dedicated to ChatGPT compared to past chatbots, such as Tay. While Tay didn't sustain media attention due to its technology limitations, it appears that within the first four months since its release date ChatGPT has been able to not only retain media interest but has had increased media coverage month over month which may be in part attributed to ChatGPT's more advanced technological state.

Example of Analysis of ChatGPT Coverage – Government Legislation and Regulation

This work is part of a larger study examining media rhetoric of ChatGPT and as such provides initial findings pertaining to media discourse covering government legislation and regulation in relation to ChatGPT. After scanning the headlines and subheadings of the 354 included media articles pertaining to ChatGPT across media outlets, a total of 20 articles (5.7%) were found to contain reference to government regulation or legislation of ChatGPT. Total number of articles pertaining to legislation and regulation (0 articles in November, 3 in December, 2 in January, 5 in February, and 10 in March) tended to follow the same increased trend in coverage as total included articles pertaining to ChatGPT.

CONCLUSION

Our data shows and provides evidence for a steady rise in media coverage of ChatGPT from late 2022 to early 2023 across our five selected media outlets. By comparison with another innovative leap in technology, Tay's coverage in 2016 never reached a similar level of hype within the same media outlets. Alarming, only 5.7% of the articles in the dataset prominently profiled the issues of government regulation/legislative oversight of ChatGPT in their headlines. This paper calls for further investigations of legacy news and social media coverage of ChatGPT in LIS, highlighting its importance to understanding public perception of such innovative information systems and the need for further public oversight and regulation of AI technologies.

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